
MIDTERM EXAM 1 (FORM A)

NAME: SOLUTIONS

ID N°: _____

INSTRUCTIONS

- This is a closed-book exam. You may not use your notes, textbook or any outside resources to assist you.
- Write your answer to each question in the space assigned to each problem. If you want the instructor to read the scratch working page for the answer to a specific problem, explicitly write that on the space assigned to said problem.
- Always remember to show your work. An answer without a proper justification will earn little to no credit, even if it is correct.
- Circle or box your final answer to each question.
- Make sure that your answers are organized and legible. Use a dark pencil or a pen. Any work that is hard or impossible to read will not be graded.
- In this exam, the use of determinants is not allowed. We will allow determinants once we cover them in this course.
- The use of calculators of any type is not allowed.

This exam consists of four problems, and your total score will be out of $10 + 10 + 10 + 10 = 40$ points.

Good Luck!

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Problem 1 (2 points per part). On each of the following questions, **completely fill in** the circle next to the correct answer. Each question only has one correct answer. You do not need to justify your choices.

(1) If A is an $m \times n$ matrix, the rank r of A always satisfies:

- ☐ $r \leq m$ (but not necessarily $r \leq n$).
- ☐ $r \leq n$ (but not necessarily $r \leq m$).
- ☒ $r \leq n$ and $r \leq m$.
- ☐ None of the above.

(2) A system of linear equations is called *consistent* if:

- ☐ It has exactly one solution.
- ☐ It has infinitely many solutions.
- ☐ It has no solutions.
- ☒ It has at least one solution.

(3) If $A\mathbf{x} = \mathbf{b}$ is a *non-homogeneous* linear system and $\mathbf{x}_1, \mathbf{x}_2$ are solutions to it, then:

- ☐ $\mathbf{x}_1 + \mathbf{x}_2$ is a solution to $A\mathbf{x} = \mathbf{b}$.
- ☐ $\mathbf{x}_1 - \mathbf{x}_2$ is a solution to $A\mathbf{x} = \mathbf{b}$.
- ☐ $\mathbf{x}_1 + \mathbf{x}_2$ is a solution to $A\mathbf{x} = \mathbf{0}$.
- ☒ $\mathbf{x}_1 - \mathbf{x}_2$ is a solution to $A\mathbf{x} = \mathbf{0}$.

(4) Given the matrices $X = \begin{bmatrix} 1 & 0 & 2 \end{bmatrix}$ and $Y = \begin{bmatrix} 2 & 1 & 3 \end{bmatrix}^T$, we have:

- ☐ The matrices $5X + Y$, $X(2Y)$ and $4Y^T$ exist.
- ☒ The matrices XY , YX and $2X + 5Y^T$ exist.
- ☐ The matrices $Y(3X^T)$, $(X + Y^T)^T$ and Y^2 exist.
- ☐ The matrices $X^T X$, $-2X + Y$ and YX^2 exist.

(5) A homogeneous system of equations $A\mathbf{x} = \mathbf{0}$ has 20 variables and 8 equations. If the augmented matrix $[A \mid \mathbf{0}]$ has rank 5, then:

- ☐ The system is consistent regardless of A , and its general solution involves 12 parameters.
- ☒ The system is consistent regardless of A , and its general solution involves 15 parameters.
- ☐ There are matrices A for which the system is inconsistent. However, when the system is consistent, its general solution involves 12 parameters.
- ☐ There are matrices A for which the system is inconsistent. However, when the system is consistent, its general solution involves 15 parameters.

Problem 2 (10 points). Suppose A and B are 3×2 matrices.

- (a) (6 points) Find, in terms of A and B , the matrix X that satisfies

$$5(X + 3A^T)^T = 2X^T - B.$$

What is the size of X ?

$$5(X + 3A^T)^T = 2X^T - B \Rightarrow 5(X^T + 3A) = 2X^T - B \Rightarrow 5X^T + 15A = 2X^T - B$$

$$\Rightarrow 3X^T = -15A - B$$

$$\Rightarrow X^T = -\frac{1}{3}(15A + B)$$

$$\Rightarrow \boxed{X = -\frac{1}{3}(15A^T + B^T)}$$

The matrix X has size 2×3 because A^T and B^T are 2×3 matrices and X is the result of adding scalar multiples of A^T and B^T .

- (b) (4 points) Calculate the matrix $C^2 - 7C + 14I_2$, where

$$C = \begin{bmatrix} 3 & -2 \\ 1 & 4 \end{bmatrix}.$$

$$C^2 - 7C + 14I_2 = \begin{bmatrix} 3 & -2 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ 1 & 4 \end{bmatrix} - 7 \begin{bmatrix} 3 & -2 \\ 1 & 4 \end{bmatrix} + 14 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & -14 \\ 7 & 14 \end{bmatrix} + \begin{bmatrix} -21 & 14 \\ -7 & -28 \end{bmatrix} + \begin{bmatrix} 14 & 0 \\ 0 & 14 \end{bmatrix}$$

$$= \boxed{\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}}$$

Problem 3 (10 points). Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 5 & 10 & -4 \\ 0 & 1 & 3 \end{bmatrix}.$$

(a) (6 points) Using elementary row operations, find the inverse A^{-1} of A .

$$\begin{aligned} \left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 5 & 10 & -4 & 0 & 1 & 0 \\ 0 & 1 & 3 & 0 & 0 & 1 \end{array} \right] &\xrightarrow{R_2 - 5R_1} \left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -5 & 1 & 0 \\ 0 & 1 & 3 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & 1 & 3 & 0 & 0 & 1 \\ 0 & 0 & 1 & -5 & 1 & 0 \end{array} \right] \\ &\xrightarrow{\substack{R_1 - 2R_2 \\ R_1 + R_3}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -4 & 1 & 0 \\ 0 & 1 & 0 & 15 & -3 & 1 \\ 0 & 0 & 1 & -5 & 1 & 0 \end{array} \right] \xrightarrow{R_1 - 2R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -34 & 7 & -2 \\ 0 & 1 & 0 & 15 & -3 & 1 \\ 0 & 0 & 1 & -5 & 1 & 0 \end{array} \right] \end{aligned}$$

So A is indeed invertible and

$$A^{-1} = \begin{bmatrix} -34 & 7 & -2 \\ 15 & -3 & 1 \\ -5 & 1 & 0 \end{bmatrix}$$

(b) (4 points) Use the previous part to find the solutions to the system

$$\begin{cases} x + 2y - z = 0 \\ 5x + 10y - 4z = -1 \\ y + 3z = 3 \end{cases}$$

The matrix form of the system is $A\vec{x} = \vec{b}$, $\vec{b} = \begin{bmatrix} 0 \\ -1 \\ 3 \end{bmatrix}$ and $\vec{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$. Thus, because the matrix A is invertible, the system is consistent with unique solution

$$\vec{x} = A^{-1}\vec{b} = \begin{bmatrix} -34 & 7 & -2 \\ 15 & -3 & 1 \\ -5 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} -13 \\ 6 \\ -1 \end{bmatrix}$$

Problem 4 (10 points). Consider the system of linear equations $A\mathbf{x} = \mathbf{b}$ whose augmented matrix is:

$$[A|\mathbf{b}] = \left[\begin{array}{ccc|c} 1 & 0 & 6 & a \\ 3 & 1 & 5 & b \\ -4 & -2 & 2 & c \end{array} \right].$$

- (a) (2 points) What conditions do a , b and c have to satisfy in order for the system $A\mathbf{x} = \mathbf{b}$ to be homogeneous?

The system is homogeneous if and only if the constant terms are all zero:

$$a=0, b=0, c=0$$

- (b) (4 points) Using the Gaussian algorithm, show that the system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if ~~$a=0, b=0$~~ . $c=2a-2b$

$$\left[\begin{array}{ccc|c} 1 & 0 & 6 & a \\ 3 & 1 & 5 & b \\ -4 & -2 & 2 & c \end{array} \right] \xrightarrow{\substack{R_2-3R_1 \\ R_3+4R_1}} \left[\begin{array}{ccc|c} 1 & 0 & 6 & a \\ 0 & 1 & -13 & b-3a \\ 0 & -2 & 26 & c+4a \end{array} \right] \xrightarrow{R_3+2R_2} \left[\begin{array}{ccc|c} 1 & 0 & 6 & a \\ 0 & 1 & -13 & b-3a \\ 0 & 0 & 0 & c-2a+2b \end{array} \right]$$

The last matrix is in row-echelon form, so the system is consistent if and only if it does not contain an equation of the form $0=\text{nonzero}$. Thus

$$\text{The system is consistent} \Leftrightarrow c-2a+2b=0 \Leftrightarrow c=2a-2b$$

(c) (4 points) Calculate a set of basic solutions to the system $A\mathbf{x} = \mathbf{0}$.

(**Hint:** Recycle the calculations from the previous part)

From the previous part, setting $a=b=c=0$ we see that the original system is equivalent to

$$\begin{cases} x + 6z = 0 \\ y - 13z = 0 \end{cases} \Rightarrow \text{The general solution is} \\ x = -6t, \quad y = 13t, \quad z = t, \quad t \text{ arbitrary} \\ \left(\text{i.e. } \vec{x} = t \begin{bmatrix} -6 \\ 13 \\ 1 \end{bmatrix} \right)$$

$\uparrow \quad \uparrow \quad \uparrow$
 Leading
 variables $z=t$

A set of basic solutions is just

$$\boxed{\vec{x}_1 = \begin{bmatrix} -6 \\ 13 \\ 1 \end{bmatrix}}$$

Scratch working page

Scratch working page